COHOMOLOGY GAP FORCES A SPLIT ON REGULAR SCHEMES

THE STACKS PROJECT AUTHORS

Allowed standard prerequisites (editorial).

- Ext vanishing. lemma-Ext-0-regular L26471 (carded this batch). Kills the connecting morphism delta.
- Split criterion. derived-lemma-split (cross-chapter, named). Vanishing delta splits the triangle.

Source reference. Original reference 05QT: focused-seeds/geometry/S9-eqv-split-complex-regular/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 08J5: focused-seeds/geometry/S9-eqv-split-complex-regular/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0FZ3: sources/stacks-equiv/pinned-original-source.tex, source lines 1862–1862. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let X be a regular Noetherian scheme of dimension $d < \infty$. Let $K \in D^b_{Coh}(\mathcal{O}_X)$ and $a \in \mathbf{Z}$. If $H^i(K) = 0$ for $a < i < a + d$, then $K = \tau_{\leq a}K \oplus \tau_{\geq a+d}K$.*

Proof. We have $\tau_{\leq a}K = \tau_{\leq a+d-1}K$ by the assumed vanishing of cohomology sheaves. By Derived Categories, Remark 08J5 we have a distinguished triangle

$$\tau_{\leq a}K \to K \to \tau_{\geq a+d}K \xrightarrow{\delta} (\tau_{\leq a}K)[1]$$

By Derived Categories, Lemma 05QT it suffices to show that the morphism δ is zero. This follows from Lemma 0FZ3. □

Date: stacks-project:equiv.tex:local-snapshot-20261012; GNU FDL 1.2 or later.